

Nutrition & Wellness During Sarcoma Treatment

Eating Well, Staying Strong, and Supporting Recovery

Evidence-based guidance on nutrition, physical activity, and wellbeing throughout sarcoma diagnosis, treatment, and survivorship.

Sarcoma Hub | Wellness Series | Document 5 of 6

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Why Nutrition Matters in Sarcoma Care

Good nutrition during cancer treatment helps you tolerate chemotherapy and radiation better, recover faster from surgery, maintain muscle mass and strength, and support your immune system. Sarcoma patients face particular challenges because many treatments — including doxorubicin, ifosfamide, and limb surgery — significantly affect appetite, weight, and physical capacity.

***IMPORTANT:** Before making significant changes to your diet or starting supplements, discuss with your oncologist or a registered oncology dietitian. Some supplements can interfere with chemotherapy effectiveness.*

General Nutritional Goals During Treatment

The primary nutritional goals during active treatment are different from general healthy eating advice. Rather than restricting calories, the focus is on:

- **Maintaining weight and muscle mass** — unintentional weight loss significantly worsens treatment tolerance and outcomes.
- **Adequate protein intake** — essential for tissue repair and immune function. Aim for 1.2-1.5g of protein per kg of body weight per day during active treatment.
- **Staying well hydrated** — especially important with ifosfamide (which requires aggressive hydration to prevent bladder toxicity) and during fever or infection.
- **Managing symptoms** — tailor food choices to minimise nausea, mouth sores, constipation, or diarrhoea depending on your treatment.
- **Micronutrient adequacy** — a varied diet rich in vegetables, fruits, whole grains, and lean protein usually provides sufficient vitamins and minerals.

Eating Through Common Treatment Side Effects

Side Effect	Nutritional strategies
Nausea / vomiting	Eat small, frequent meals every 2-3 hours. Cold or room-temperature foods are better tolerated. Ginger (tea, capsules, sweets) has evidence for chemo nausea. Avoid fatty, spicy, or strongly odorous foods. Stay upright for 30 min after eating.
Mouth sores (mucositis)	Soft, moist foods: yoghurt, scrambled eggs, mashed potato, smoothies, porridge. Rinse mouth with salt water or sodium bicarbonate solution 4-6x/day. Avoid alcohol mouthwashes. Cool foods may soothe. Avoid acidic foods (citrus, tomato, vinegar).
Appetite loss	Eat by the clock, not by hunger. Fortify meals with calorie-dense additions: nut butters, olive oil, full-fat dairy, avocado, eggs. Nutritional supplement drinks (e.g. Ensure, Fortisip) can help bridge gaps. Eat your largest meal when appetite is best (often morning).
Constipation	Increase fluid intake. Gentle soluble fibre: oats, cooked vegetables, fruit. Prunes and prune juice. Light physical movement if possible. Discuss laxatives with your team — do not strain.
Diarrhoea	BRAT-adjacent foods: bananas, white rice, plain pasta, boiled chicken, toast. Avoid high-fibre, fatty, or spicy foods during acute episodes. Oral rehydration solutions to replace electrolytes. Inform your team if severe or persistent.
Taste changes (dysgeusia)	Common with chemotherapy. Metallic taste: use plastic cutlery, marinate meats in fruit juices. Experiment with flavour enhancers: herbs, lemon, vinegar. Zinc supplementation may help — discuss with your team. Taste usually returns after treatment ends.

High-Protein Foods for Muscle Maintenance

Muscle loss (sarcopenia) is a significant risk during sarcoma treatment, particularly with limb surgery or prolonged chemotherapy. Prioritise these protein-rich foods:

Food	Protein	Tips for treatment
Greek yoghurt (200g)	~18-20g	Easy to eat, probiotic benefit, add to smoothies
Eggs (2 large)	~12g	Versatile, soft when scrambled, excellent amino acid profile
Chicken breast (100g cooked)	~31g	Lean, adaptable; poach or steam for easy texture
Salmon / oily fish (100g)	~25g	Also provides omega-3 anti-inflammatory fats
Lentils / legumes (200g cooked)	~18g	Plant-based, high in iron and fibre
Tofu (150g firm)	~16g	Neutral flavour; absorbs sauces; soft when silken
Cottage cheese (200g)	~26g	High protein, mild taste, easy to eat cold
Protein supplement shakes	20-30g per serving	Useful when food intake is very low

Physical Activity During Treatment

Exercise during cancer treatment is now strongly supported by evidence. It reduces fatigue, improves mood, preserves muscle mass, and may improve treatment tolerance. Even gentle activity makes a difference.

- **Always consult your medical team** before starting any exercise programme during active treatment.
- **Walking** — even 10-15 minutes daily is beneficial. Start gently and build gradually.
- **Resistance/strength exercises** — with physiotherapist guidance, especially important after limb surgery to rebuild function.
- **Avoid exercise during** neutropenic periods (when immune cells are low), within 24-48 hours of chemotherapy, or if you have bone metastases without specialist clearance.
- **Fatigue is not a reason to rest completely** — rest and gentle activity must be balanced. Prolonged bed rest worsens fatigue.

Supplements: What the Evidence Says

Supplement	Evidence	Safety Note
Vitamin D	Many cancer patients are deficient. Supplementation (1,000-2,000 IU/day) is generally safe and may support bone health and immunity. Check levels with a blood test.	Low risk; discuss dose with team
Omega-3 fatty acids	Some evidence for reducing inflammation and preserving muscle mass during chemotherapy. Fish oil supplements 1-2g EPA/DHA per day.	Generally safe; may affect bleeding — discuss with surgeon pre-op

Antioxidants (high-dose)	High-dose antioxidant supplements (e.g. vitamin C, vitamin E) during chemotherapy may theoretically reduce treatment efficacy. Evidence is unclear.	AVOID high-dose antioxidants during active chemo without oncologist approval
Zinc	May help with taste changes and immune function. Discuss dosing with your team.	Low risk at moderate doses
Probiotics	May reduce chemotherapy-related diarrhoea. Some evidence but not standard of care. Avoid in severely immunocompromised patients.	Discuss with oncologist; avoid in neutropenia
St John's Wort	Interacts significantly with many medications including chemotherapy agents. AVOID during treatment.	DO NOT TAKE during cancer treatment

This guide provides general nutritional information only. Individual needs vary significantly by treatment type, sarcoma subtype, and patient health status. Please work with a registered oncology dietitian for personalised guidance. Sarcoma Hub — sarcomahub.org